

1. A Transformer Network, like its predecessors RNNs, GRUs and LSTMs, can process information one word at a time. (Sequential architecture).

1 point

- ☐ False
- ☐ True

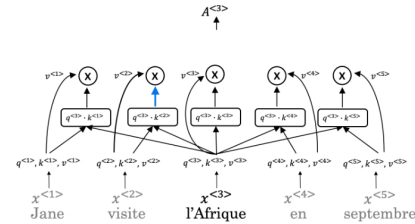
2. The major innovation of the transformer architecture is combining the use of LSTMs and RNN sequential processing.

1 point

- ☐ True
- ☐ False

3. What are the key inputs to computing the attention value for each word?

1 point



- ☐ The key inputs to computing the attention value for each word are called the query, knowledge, and vector.
- ☐ The key inputs to computing the attention value for each word are called the quotation, knowledge, and value.
- ☐ The key inputs to computing the attention value for each word are called the query, key, and value.
- ☐ The key inputs to computing the attention value for each word are called the quotation, key, and vector.

4. Which of the following correctly represents Attention?

1 point

- ☐ $A(Q, K, V) = \left(\frac{\exp(\text{qk}^{(-i)})}{\sum_j \exp(\text{qk}^{(-j)})} \right) * V^{(-i)}$
- ☐ $A(Q, K, V) = \sum_i \left(\frac{\exp(\text{qk}^{(-i)})}{\sum_j \exp(\text{qk}^{(-j)})} \right) * \sum_i v^i$
- ☐ $A(Q, K, V) = \sum_i \left(\frac{\exp(\text{qk}^{(-i)})}{\sum_j \exp(\text{qk}^{(-j)})} \right) * V^{(-i)}$
- ☐ $A(Q, K, V) = \sum_i \left(\frac{\exp(\text{qkv}^{(-i)})}{\sum_j \exp(\text{qkv}^{(-j)})} \right) * K^{(-i)}$

5. Are the following statements true regarding Query (Q), Key (K) and Value (V) ?

1 point

Q = interesting questions about the words in a sentence

K = qualities of words given a Q

V = specific representations of words given a Q

- ☐ False
- ☐ True

$$\text{Attention}(W_i^Q Q, W_i^K K, W_i^V V)$$

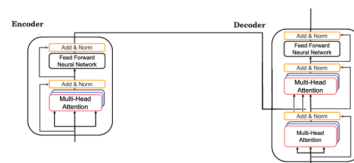
1 point

6. i here represents the computed attention weight matrix associated with the i th "word" in a sentence.

- ☐ False
- ☐ True

7. Following is the architecture within a Transformer Network (without displaying positional encoding and output layers(s)).

1 point

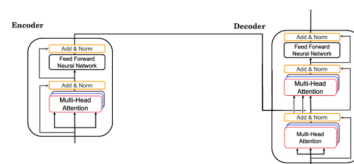


What is generated from the output of the Decoder's first block of Multi-Head Attention?

- ☐ K
- ☐ Q
- ☐ V

8. Following is the architecture within a Transformer Network (without displaying positional encoding and output layers(s)).

1 point



The output of the decoder block contains a softmax layer followed by a linear layer to predict the next word one word at a time.

- ☐ True
- ☐ False

9. Why is positional encoding important in the translation process? (Check all that apply)

1 point

- ☒ Position and word order are essential in sentence construction of any language.
- ☐ It helps to locate every word within a sentence.

- ☒ It is used in CNN and works well there.
- ☐ Providing extra information to our model.

10. Which of these is a good criterion for a good positional encoding algorithm?

1 point

- ☐ It should output a common encoding for each time-step (word's position in a sentence).
- ☒ The algorithm should be able to generalize to longer sentences.
- ☐ Distance between any two time-steps should be inconsistent for all sentence lengths.
- ☐ It must be nondeterministic.